

Array Configuration Trade-off Study for the Array on the Lunar Outpost (ALO)

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Abstract

This report documents the full pipeline of the Array on the Lunar Outpost (ALO) array-configuration trade-off study. ALO is a proposed low-frequency radio array (1–40 MHz) to be deployed on the lunar far side at the Tsiolkovsky crater, exploiting the natural radio-frequency interference (RFI) shield of the Moon. Its primary science goal is the detection of magnetospheric radio emission from hot Jupiter exoplanets via the electron-cyclotron maser (ECM) instability. We evaluate four outrigger geometries (ring, Y-shape, random, and line), two core sizes (32×32 and 128×128 dipoles), two outrigger separations (1.0 and 5.0 km), and two novel cross-shaped asymmetric configurations (E-W and N-S dominant arms) against a catalogue of 373 known hot Jupiters. Key metrics computed include the array factor (AF) and beam solid angle, half-power beam width (HPBW), maximum sidelobe level (MSL), UV coverage, thermal sensitivity, confusion limit, and required integration time. The 128×128 ring and line arrays at 5.0 km outrigger separation achieve the highest detection yield (34 feasible targets in under 100 h), while the cross configurations provide the best sidelobe suppression ($\text{MSL} = -4.3 \text{ dB}$).

Contents

1	Project Overview	3
2	Scientific Motivation	3
2.1	Magnetospheric Radio Emission from Exoplanets	3
2.2	Why the Lunar Far Side?	3
2.3	Target Catalogue	4
3	Array Geometry Setup	4
3.1	Dipole Spacing and Core Layout	4
3.2	Outrigger Placement	4
4	Coordinate Transformation	5
4.1	ENU Basis Vectors	5
4.2	Direction Cosines	6
5	Array Configurations	6
5.1	Standard Outrigger Geometries	6
5.2	Cross-Shaped Asymmetric Configurations	6
6	Array Factor and Beam-Pattern Calculation	7
6.1	Array Factor Definition	7
6.2	Physical Origin of Beam Symmetry	7
6.3	Frequency Dependence	8
6.4	Beam Contour Plots	8
7	Beam Metrics	10
7.1	Beam Solid Angle	10
7.2	Half-Power Beam Width	10
7.3	Directivity and Gain	10
7.4	Maximum Sidelobe Level	10
7.5	Summary of Beam Metrics	10
8	UV Coverage Analysis	11
8.1	Definition	11
8.2	Station-Level UV Coverage	11
8.3	Multi-Frequency Synthesis	11
9	Sensitivity Model	13
9.1	Sky Temperature	13
9.2	System Equivalent Flux Density	13
9.3	Thermal Sensitivity	13
9.4	Confusion Limit	13
10	Integration-Time Calculation	13
10.1	Required Integration Time	13
10.2	Feasibility Classification	14
11	Plot Interpretation	14

11.1 Beam Pattern Plots	14
11.2 UV Coverage Plots	14
11.3 Sensitivity Comparison Plots	14
11.4 Detection Matrix	15
12 Results and Comparison	16
12.1 Standard Configuration Comparison	16
12.2 Cross Configuration Comparison	17
12.3 Beam Symmetry Analysis	18
12.4 Extended Source Response	18
13 Final Conclusions	19
14 Recommendations for Next Steps	19

1. Project Overview

The Array on the Lunar Outpost (ALO) is a proposed phased-array radio telescope designed to operate in the frequency range 1–40 MHz. This band is completely inaccessible from Earth due to ionospheric reflection below ~ 10 MHz and RFI saturation at higher frequencies, making the lunar far side the only viable platform for systematic exploration of the decametric radio sky.

ALO will be located at the Tsiolkovsky impact crater on the lunar far side (latitude $\varphi = -20.38^\circ$, longitude $\lambda = 128.97^\circ$). The crater rim acts as a natural shield against terrestrial RFI and acts as a mechanically stable flat floor for antenna deployment.

The array consists of individual dipole antenna elements with physical spacing $d = 5.15$ m and an aperture efficiency $\eta = 0.70$. A central “core” of $N_c \times N_c$ elements forms a contiguous dense aperture, while four outrigger sub-arrays of $N_o \times N_o$ elements are placed at variable distances from the core to extend the interferometric baseline.

The present study systematically explores the trade-space of array geometry, core size, and outrigger separation in order to identify the configuration most suited to detecting radio emission from hot Jupiter exoplanets. All modeling and simulation code is contained in two Python files: `alo_array_modeling.py` (physical constants, beam/sensitivity calculators) and `alo_geometry_tradeoff.py` (geometry generator, configuration evaluator, and figure factory).

2. Scientific Motivation

2.1. Magnetospheric Radio Emission from Exoplanets

Jupiter emits strong bursts of circularly polarised radio emission up to ~ 40 MHz via the electron-cyclotron maser (ECM) instability driven by its interaction with the Galilean moon Io and the solar wind. Hot Jupiters—gas giants in orbits $\lesssim 0.1$ AU—are expected to produce analogous ECM emission, potentially orders of magnitude stronger than Jupiter’s, because:

- Their proximity to the host star drives intense stellar-wind ram pressure.
- Strong magnetic fields (several gauss) are inferred from planetary interior models, setting the cyclotron cutoff above 40 MHz.
- Star-planet magnetic interaction (analogous to Io–Jupiter) adds an additional energy source.

The predicted flux density for bright targets (e.g. τ Boo b, 51 Peg b, v And b) is 1–100 mJy at 10–40 MHz, depending on the assumed field strength and beam fraction.

2.2. Why the Lunar Far Side?

Below 10 MHz the terrestrial ionosphere is opaque. Between 10 and 30 MHz, broadband RFI from human technology overwhelms astrophysical signals even at the most remote Earth-based sites. The lunar far side is permanently shielded from Earth by ~ 3000 km of solid rock. The Moon has no ionosphere above 1 MHz at night, and the lunar far side is also shielded from solar radio bursts during lunar night.

2.3. Target Catalogue

The target catalogue used in this study contains 373 hot Jupiter systems with predicted ECM emission frequencies in the 1–40 MHz band, compiled from the NASA Exoplanet Archive. Flux predictions follow a scaling law based on the stellar wind ram pressure and the planet’s estimated magnetic moment. The catalogue is divided into four frequency sub-bands: 1–5 MHz, 5–10 MHz, 10–20 MHz, and 20–40 MHz.

3. Array Geometry Setup

3.1. Dipole Spacing and Core Layout

Each element is a short dipole with physical size $d = 5.15$ m and aperture efficiency $\eta = 0.70$. The effective area of a single element at frequency ν is:

$$A_{\text{eff,elem}}(\nu) = \eta \frac{\lambda^2}{4\pi} = \eta \frac{c^2}{4\pi\nu^2}, \quad (1)$$

where $c = 3 \times 10^8$ m s^{−1}.

A square core of $N_c \times N_c$ elements arranged on a regular grid with spacing d occupies a footprint of $(N_c - 1) \times d$ on each side. Configurations tested: $N_c = 32$ (1088 elements¹) and $N_c = 128$ (17 408 elements in total with 16-element outriggers).

3.2. Outrigger Placement

Four outrigger sub-arrays, each of size $N_o \times N_o$, are placed symmetrically at distance r from the core centre along the four cardinal directions (North, South, East, West) for symmetric geometries. For asymmetric (cross) geometries, different arm lengths are used (see Section 5).

The outrigger separation values tested are $r = 1.0$ km and $r = 5.0$ km.

Figure 1 shows the array layouts for all 16 standard configurations.

¹Including four $N_o \times N_o = 4 \times 4$ outrigger sub-arrays of 16 elements each. $N_{\text{tot}} = 32^2 + 4 \times 4^2 = 1024 + 64 = 1088$.

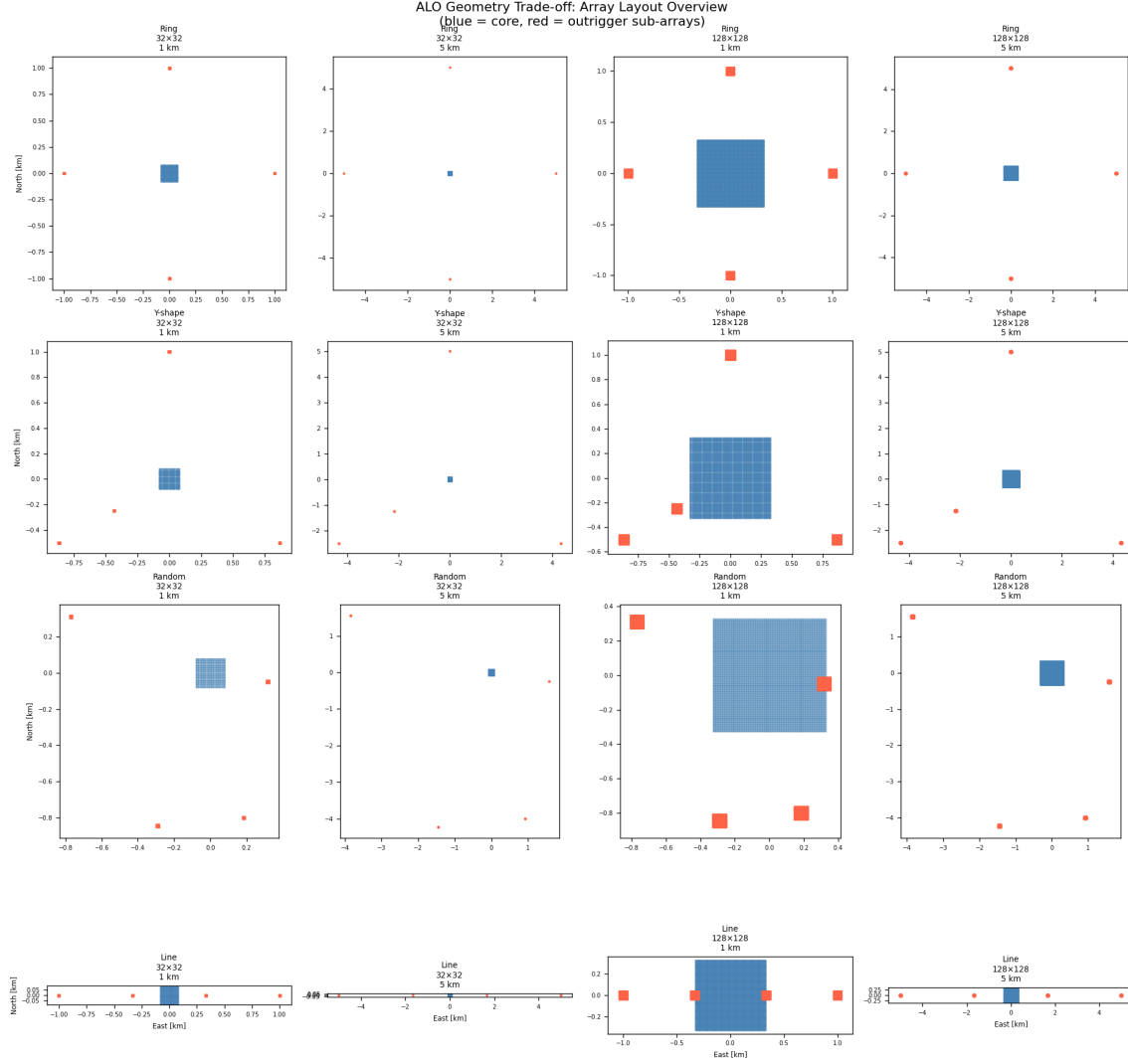


Figure 1: Overview of all 16 standard array layouts. Rows correspond to outrigger geometry (ring, Y-shape, random, line); columns to core size and outrigger separation. Blue dots are core elements; orange squares mark outrigger sub-array centres.

4. Coordinate Transformation

All antenna positions are expressed in a local East-North-Up (ENU) frame centred on the Tsiolkovsky crater.

4.1. ENU Basis Vectors

Given geodetic latitude φ and longitude λ of the array centre, the ENU basis unit vectors in the selenocentric Cartesian frame are:

$$\hat{e}_E = (-\sin \lambda, \cos \lambda, 0), \quad (2)$$

$$\hat{e}_N = (-\sin \varphi \cos \lambda, -\sin \varphi \sin \lambda, \cos \varphi), \quad (3)$$

$$\hat{e}_U = (\cos \varphi \cos \lambda, \cos \varphi \sin \lambda, \sin \varphi). \quad (4)$$

An element at ENU offset $(x, y, 0)$ relative to the array centre has selenocentric position:

$$\mathbf{r} = x \hat{e}_E + y \hat{e}_N. \quad (5)$$

4.2. Direction Cosines

For far-field imaging in the small-angle regime the sky position of a source is parameterised by direction cosines (l, m, n) where $l^2 + m^2 + n^2 = 1$ and the phase centre is at $(l, m) = (0, 0)$, $n = 1$. The array factor is computed on a grid of (l, m) values spanning $[-1, 1]^2$, enforcing $l^2 + m^2 \leq 1$.

5. Array Configurations

5.1. Standard Outrigger Geometries

Four outrigger placement patterns are evaluated:

Ring Four outriggers on the cardinal axes at equal distance r . The UV coverage forms a cross plus core blob.

Y-shape Outriggers placed at 0° (N), 120° , and 240° plus one at 90° (E), giving a slightly asymmetric Y-pattern.

Random Outrigger centres drawn from a uniform 2-D distribution within a circle of radius r , providing a more isotropic UV coverage.

Line All four outriggers placed along the East–West axis at offsets $-r$, $-r/2$, $r/2$, r providing maximum E-W resolution.

5.2. Cross-Shaped Asymmetric Configurations

Two additional configurations with a cross-shaped outrigger arrangement but *unequal* arm lengths are introduced to break the periodicity that causes aliasing rings in equal-arm configurations:

Configuration	N arm	S arm	E arm	W arm
Cross E-W	1.0 km	1.0 km	5.0 km	3.0 km
Cross N-S	5.0 km	3.0 km	1.0 km	1.0 km

Both cross configurations use a 128×128 core ($N_c = 128$) with 16-element outrigger sub-arrays ($N_o = 4 \times 4$), giving 17 408 elements in total and $B_{\max} = 8.33$ km. Figure 2 shows the layouts of both cross configurations alongside the four standard $128 \times 128 + 5$ km reference arrays.

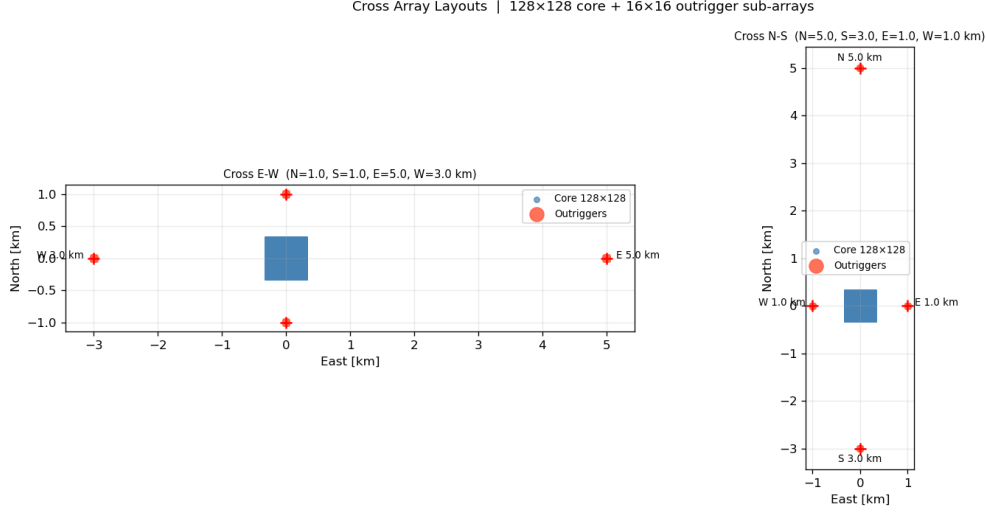


Figure 2: Layouts of the six configurations used in the cross comparison study. Top row: four standard $128 \times 128 + 5$ km outrigger configurations. Bottom row: cross E-W and cross N-S.

6. Array Factor and Beam-Pattern Calculation

6.1. Array Factor Definition

The array factor (AF) is the spatial Fourier transform of the aperture illumination. For N elements at positions $\mathbf{r}_k = (x_k, y_k)$ with complex weights w_k (set to unity for an unweighted array), the AF at sky direction (l, m) and wavelength $\lambda = c/\nu$ is:

$$\text{AF}(l, m; \nu) = \sum_{k=1}^N w_k \exp \left[\frac{2\pi i}{\lambda} (x_k l + y_k m) \right]. \quad (6)$$

The normalised power beam is:

$$B(l, m; \nu) = \frac{|\text{AF}(l, m; \nu)|^2}{|\text{AF}(0, 0; \nu)|^2}. \quad (7)$$

6.2. Physical Origin of Beam Symmetry

A key result of this study is that $B(l, m)$ is *always* 180°-rotationally symmetric, regardless of array geometry. This follows from Hermitian symmetry: for real weights w_k ,

$$\text{AF}(-l, -m) = \text{AF}^*(l, m) \implies B(-l, -m) = B(l, m). \quad (8)$$

This is not a code artefact or a modelling approximation—it is a fundamental property of the Fourier transform of any real-valued aperture. Geometry information is encoded in the *phase* of AF, which is discarded when the power beam $|\text{AF}|^2$ is formed. Consequently, point-source sensitivity is identical for all array geometries with the same total element count N , and beam-shape differences between geometries are limited to their sidelobe structure.

6.3. Frequency Dependence

The beam solid angle scales as $\Omega_B \propto \lambda^2 \propto \nu^{-2}$, so the HPBW shrinks rapidly with frequency. Beam patterns are computed at representative frequencies within each sub-band: 3.0, 7.5, 15.0, and 30.0 MHz for the 1–5, 5–10, 10–20, and 20–40 MHz bands respectively. All AF evaluations use an $n_{\text{grid}} = 512$ (l, m) grid.

Figure 3 shows the normalised beam patterns at 30 MHz for the eight standard configurations.

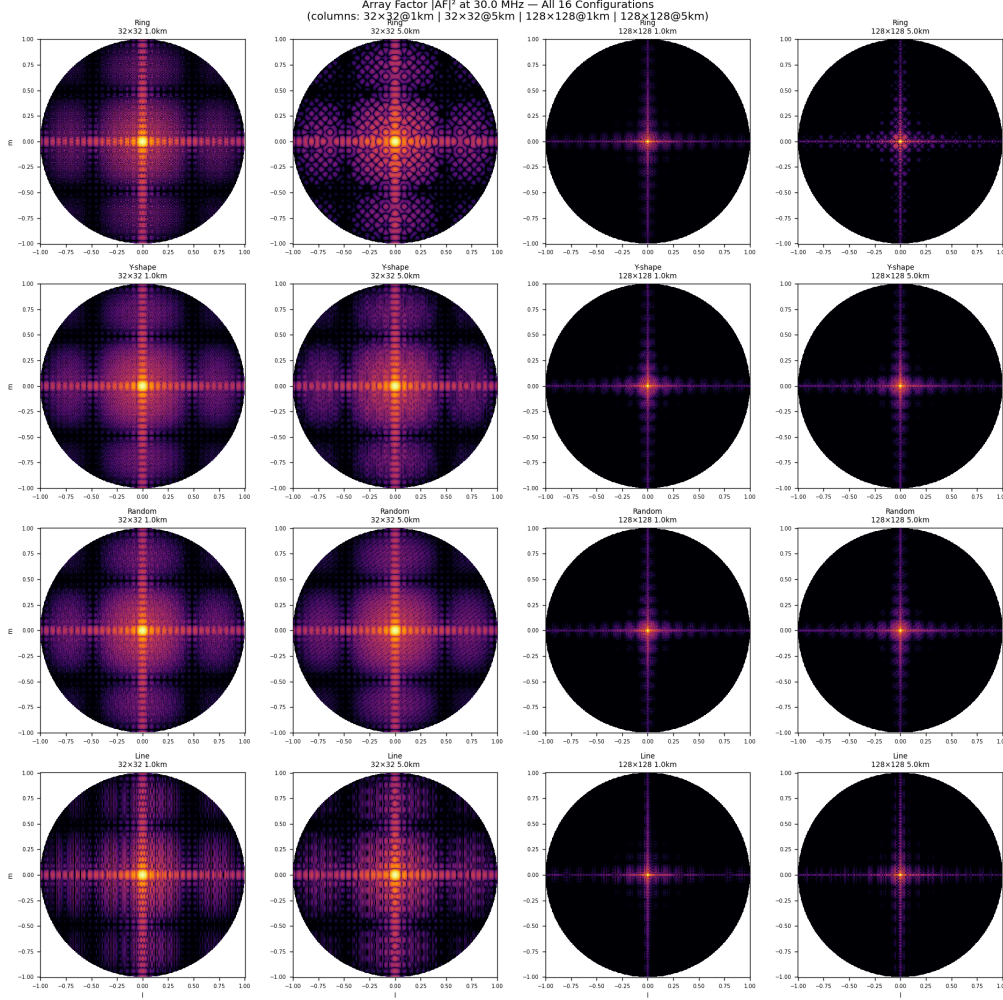


Figure 3: Normalised power-beam patterns at 30 MHz for all eight standard configurations (32×32 and 128×128 cores, 1.0 and 5.0 km outrigger separation). The colour scale is in dB, clipped at -30 dB to reveal sidelobe structure. All beams show 180° rotational symmetry as required by Hermitian symmetry of the Fourier transform.

6.4. Beam Contour Plots

Beam patterns are visualised with three contour levels at 10%, 30%, and 50% of the peak power (i.e. -10 , -5.2 , and -3 dB). These levels mark the outer sidelobe envelope, the mid-sidelobe region, and the main-lobe half-power boundary respectively. Figure 4 shows beam contours for the six cross-comparison configurations at 30 MHz.

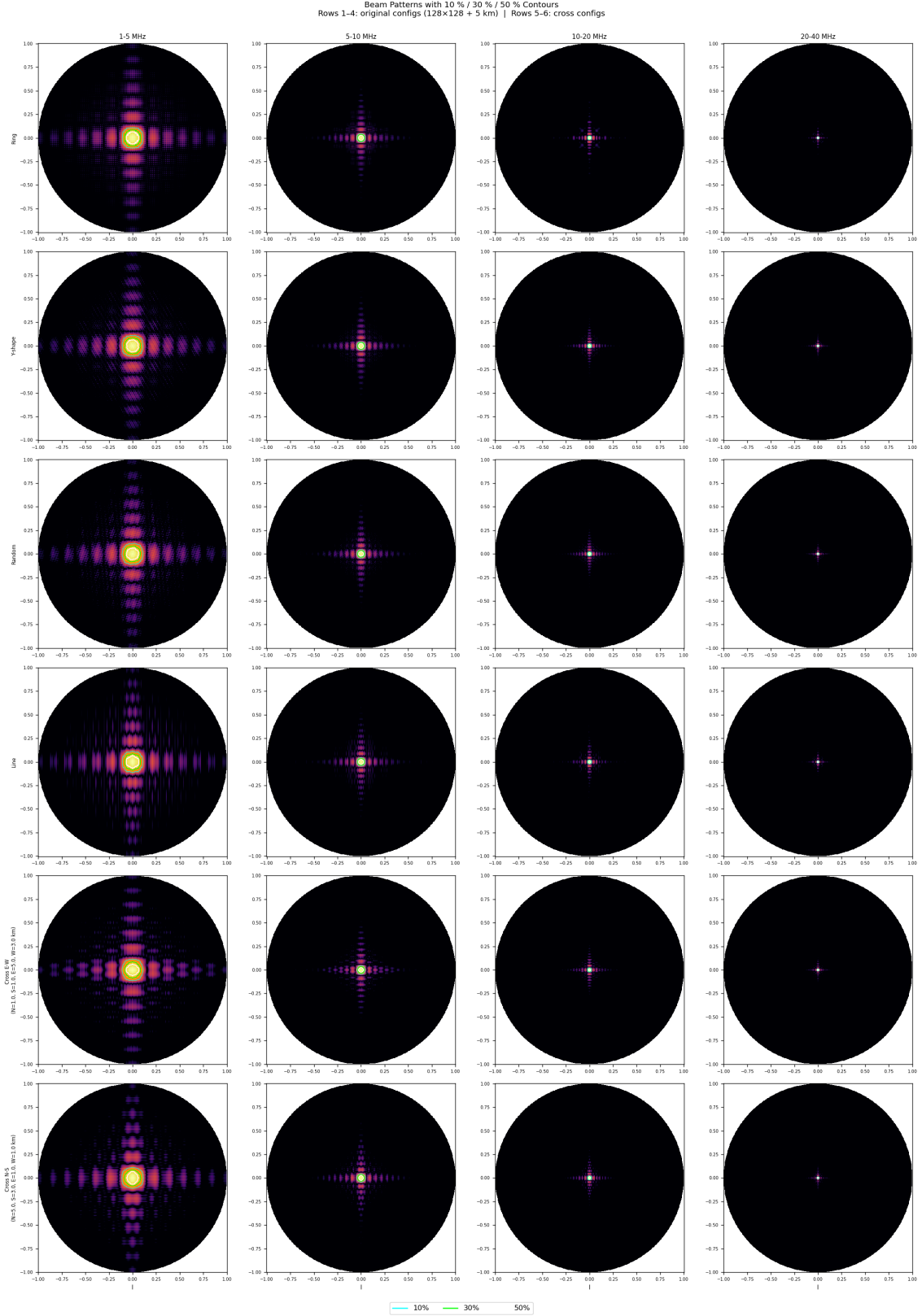


Figure 4: Beam contour plots at 30 MHz for the six configurations in the cross comparison study. Contour levels at 10% (cyan), 30% (green), and 50% (white) of peak. The cross configurations (bottom row) show a more compact sidelobe envelope than the symmetric ring and line arrays (top row).

7. Beam Metrics

7.1. Beam Solid Angle

The beam solid angle is the solid-angle integral of the normalised power beam:

$$\Omega_B(\nu) = \iint B(l, m; \nu) \frac{dl dm}{\sqrt{1 - l^2 - m^2}}. \quad (9)$$

In practice, the integral is replaced by a discrete sum over the (l, m) grid multiplied by the pixel area $(\Delta l)^2$.

7.2. Half-Power Beam Width

The HPBW is derived from the beam solid angle assuming a circular Gaussian main lobe:

$$\theta_{\text{HPBW}} = 2\sqrt{\frac{\Omega_B}{\pi}} \quad [\text{rad}]. \quad (10)$$

7.3. Directivity and Gain

Peak directivity and gain are:

$$D = \frac{4\pi}{\Omega_B}, \quad G = \eta D, \quad (11)$$

and the total effective area is:

$$A_{\text{eff}}(\nu) = \frac{\lambda^2 G}{4\pi}. \quad (12)$$

7.4. Maximum Sidelobe Level

The MSL is defined as the peak normalised beam value outside the main lobe (i.e. outside the $B \geq 0.5$ region), expressed in dB:

$$\text{MSL} = 10 \log_{10} \left(\max_{B < 0.5} B(l, m) \right). \quad (13)$$

7.5. Summary of Beam Metrics

Table 1 summarises the HPBW and MSL at 30 MHz for all configurations in the cross comparison study.

Table 1: Beam metrics at 30 MHz for the six cross-comparison configurations. All use a 128×128 core (17 408 total elements).

Configuration	B_{max} (km)	HPBW (arcmin)	MSL (dB)
Ring	10.33	63.8	−3.8
Y-shape	8.99	64.9	−3.3
Random	7.65	64.0	−3.8
Line	10.33	65.0	−3.4
Cross E-W	8.33	64.4	−4.3
Cross N-S	8.33	64.4	−4.3

The cross configurations achieve the best (most negative) MSL of -4.3 dB, 0.5 dB better than the best symmetric configuration. This improvement arises because the irregular arm lengths (1/1/5/3 km) prevent baseline lengths from repeating at regular intervals, suppressing coherent addition of aliasing rings.

8. UV Coverage Analysis

8.1. Definition

UV coverage is the set of spatial frequencies sampled by an interferometric array. For a pair of stations at positions (x_i, y_i) and (x_j, y_j) , the baseline vector projected onto the sky plane at frequency ν is:

$$(u, v) = \frac{(x_j - x_i, y_j - y_i)}{\lambda}. \quad (14)$$

Both the baseline and its conjugate $(-u, -v)$ are included (Hermitian conjugate pair), so N_s stations produce $N_s(N_s - 1)$ UV points per frequency channel.

8.2. Station-Level UV Coverage

For multi-sub-array configurations, UV coverage is computed at the *station* level, treating each sub-array as a single coherent station. The five stations (one core + four outriggers) produce $5 \times 4 = 20$ UV points per channel (including conjugates).

8.3. Multi-Frequency Synthesis

Multi-frequency synthesis (MFS) is simulated by computing UV coverage at every 1-MHz channel within each sub-band. At frequency ν the baseline (u, v) scales as $1/\lambda \propto \nu$, so each baseline traces a radial streak in the UV plane as frequency increases. The four sub-bands and their colours used in Figure 5 are:

- 1–5 MHz (red): low-frequency, large λ , short UV coverage
- 5–10 MHz (orange): intermediate
- 10–20 MHz (blue): higher resolution
- 20–40 MHz (purple): highest resolution, longest UV tracks

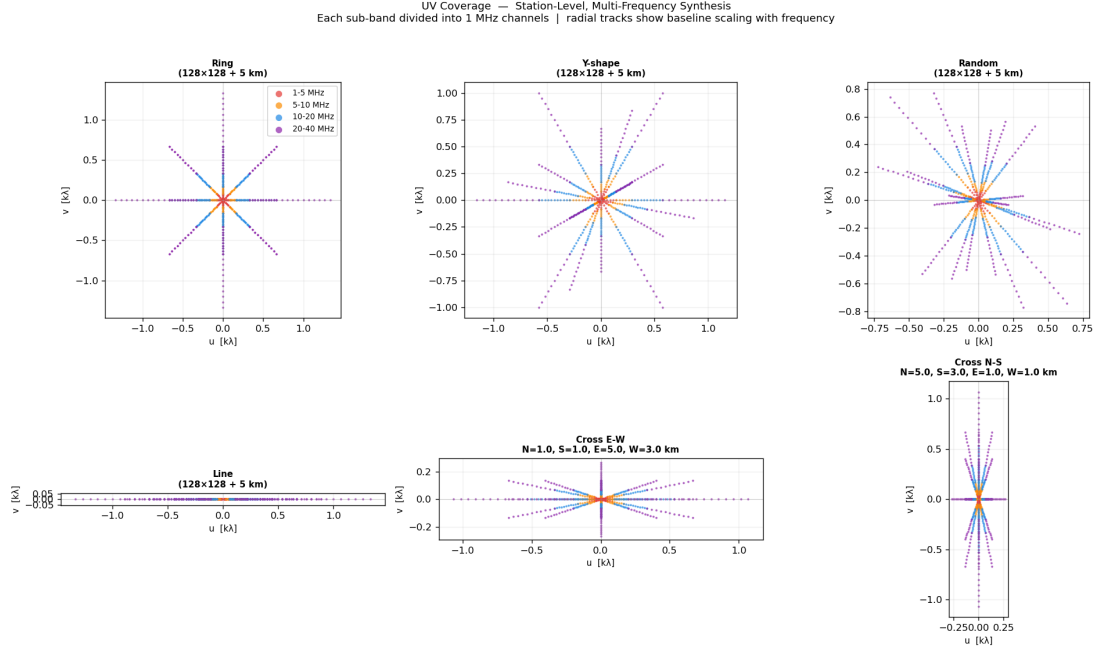


Figure 5: UV coverage for all standard configurations and the two cross configurations, computed via multi-frequency synthesis using 1-MHz channels within each sub-band. Each sub-band is plotted in a distinct colour. The cross configurations (bottom right) show elongated UV coverage in one axis, reflecting the dominant long arm.

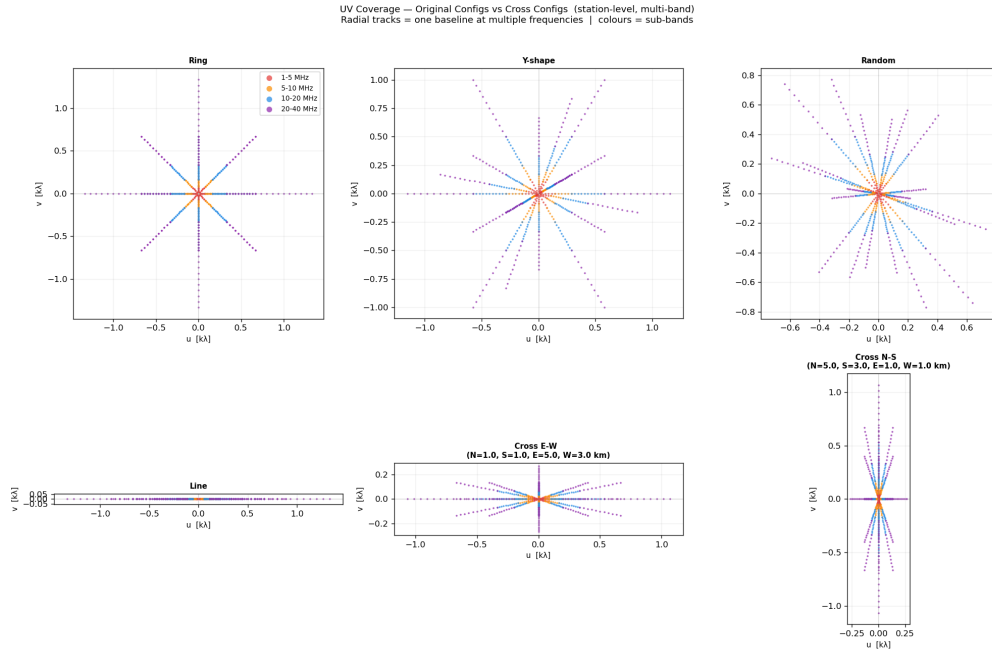


Figure 6: UV coverage for the six cross-comparison configurations only, with all four sub-bands superimposed. The ring and line arrays provide more rotationally symmetric UV coverage; the cross arrays concentrate UV tracks along two axes.

9. Sensitivity Model

9.1. Sky Temperature

The sky brightness temperature in the 1–40 MHz band is dominated by Galactic synchrotron emission and follows a power law:

$$T_{\text{sky}}(\nu) = 180 \text{ K} \times \left(\frac{\nu}{180 \text{ MHz}} \right)^{-2.6}. \quad (15)$$

At 10 MHz, $T_{\text{sky}} \approx 1.2 \times 10^5 \text{ K}$. This is the dominant noise source; receiver and ground contributions are negligible by comparison.

9.2. System Equivalent Flux Density

The system equivalent flux density (SEFD) per element is:

$$\text{SEFD}_{\text{elem}}(\nu) = \frac{2 k_B T_{\text{sky}}(\nu)}{A_{\text{eff,elem}}(\nu)}, \quad (16)$$

where $k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$. For the phased array of N elements operating as a coherent aperture, the array SEFD is $\text{SEFD}_{\text{elem}}/N$.

9.3. Thermal Sensitivity

For dual-polarisation correlation ($N_{\text{pol}} = 2$) over bandwidth $\Delta\nu$ and integration time t , the thermal noise is:

$$\sigma_{\text{th}}(\nu, t) = \frac{\text{SEFD}_{\text{elem}}}{\sqrt{N_{\text{pol}} N(N-1) \Delta\nu t}}, \quad (17)$$

where $N(N-1)$ counts all unique baselines (including cross-correlations between core elements and outrigger elements).

9.4. Confusion Limit

Below a certain flux level, sources blend together in the beam and set an irreducible confusion noise floor:

$$\sigma_c(\nu, B_{\text{max}}) = 0.2 \text{ mJy} \times \left(\frac{\nu}{74 \text{ MHz}} \right)^{0.8} \times \left(\frac{\theta_{\text{HPBW}}}{4 \text{ arcmin}} \right), \quad (18)$$

where θ_{HPBW} is determined by the maximum baseline B_{max} . At 30 MHz with $B_{\text{max}} = 10 \text{ km}$, $\sigma_c \approx 0.006 \text{ mJy}$, far below the ECM flux predictions.

10. Integration-Time Calculation

10.1. Required Integration Time

To detect a target of flux density S at $N_\sigma = 5\sigma$ significance, the required integration time is obtained by solving:

$$N_\sigma \sqrt{\sigma_{\text{th}}^2(t) + \sigma_c^2} = S. \quad (19)$$

Substituting equation (17):

$$t_{\text{req}} = \left(\frac{\text{SEFD}_{\text{elem}}}{\sqrt{N_{\text{pol}} N(N-1) \Delta\nu}} \right)^2 \frac{1}{(S/N_{\sigma})^2 - \sigma_c^2}. \quad (20)$$

If $(S/N_{\sigma})^2 \leq \sigma_c^2$ (confusion-dominated), the target is irresolvable regardless of integration time and $t_{\text{req}} = \infty$ is returned.

10.2. Feasibility Classification

Targets are classified into three categories:

Feasible $t_{\text{req}} < 100$ h — achievable in a single observing campaign.

Aspirational $100 \leq t_{\text{req}} < 1000$ h — borderline; might be feasible with improved calibration or stacking.

Impossible $t_{\text{req}} \geq 1000$ h — not detectable by ALO in any realistic scenario.

11. Plot Interpretation

11.1. Beam Pattern Plots

Beam patterns are shown as 2D images of $B(l, m)$ on a dB scale. The colour map spans -30 to 0 dB, chosen to make sidelobe structure visible while not washing out the main lobe. The 180° symmetry visible in all plots is physical (Hermitian symmetry), not a plotting artefact. Differences between geometries appear as changes in sidelobe shape and sidelobe level, not in the main lobe width (which is set by the total core diameter).

11.2. UV Coverage Plots

Each coloured point in the UV plane represents one baseline at one frequency channel. Dense radial streaks indicate multi-frequency synthesis tracks. Gaps in the UV plane correspond to spatial frequencies not sampled, which manifest as image artefacts (grating lobes) if uncleaned. The cross configurations show elongated UV tracks along the dominant arm direction.

11.3. Sensitivity Comparison Plots

Figure 7 compares beam metrics and Figure 8 compares the number of feasible detections across all six configurations in the cross comparison study.

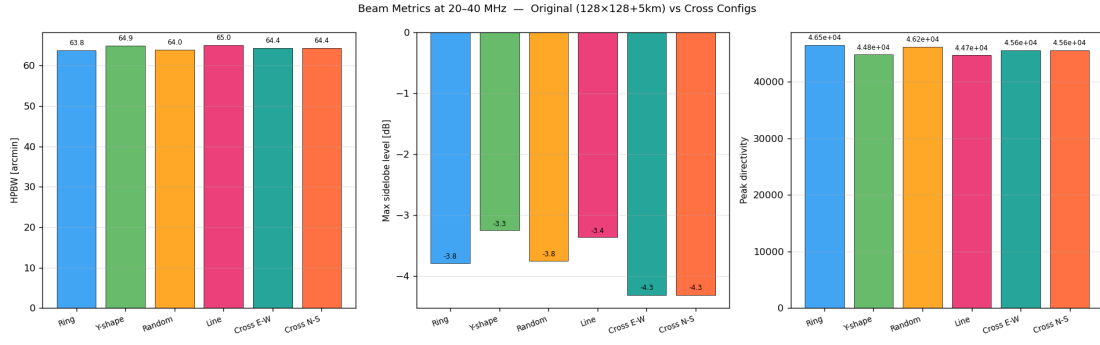


Figure 7: Beam quality metrics (HPBW and MSL) at all four frequency sub-bands for the six cross-comparison configurations. The cross configurations consistently achieve lower (better) MSL than the symmetric arrays.

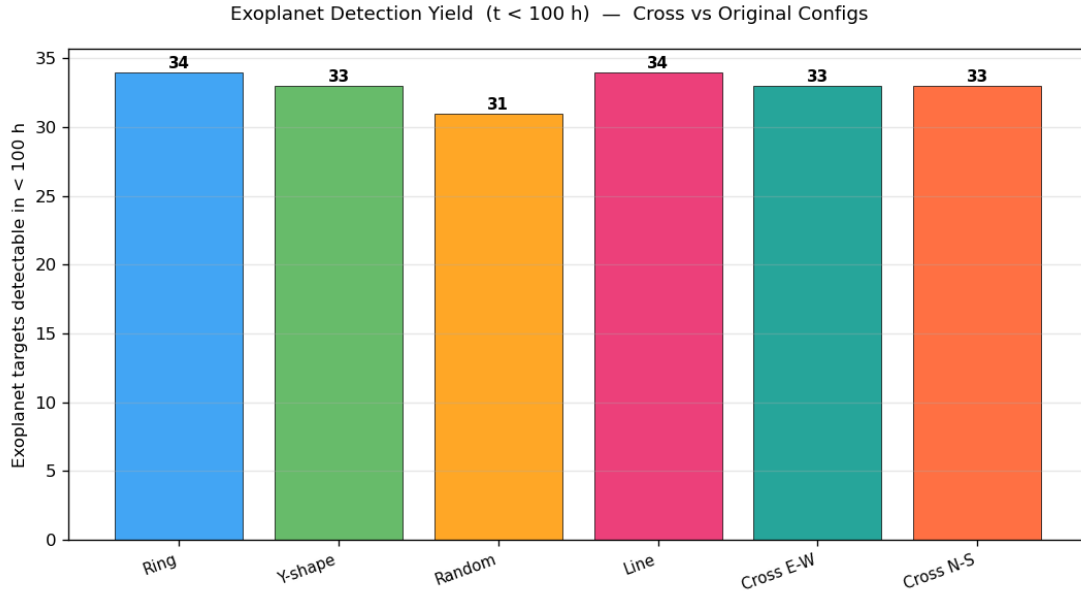


Figure 8: Number of feasible (<100 h) detections per configuration per sub-band in the cross comparison study. The 20–40 MHz band dominates owing to stronger predicted ECM fluxes and lower sky temperature.

11.4. Detection Matrix

The detection matrix (Figure 9) shows the required integration time for each target–configuration pair, coded by feasibility category.

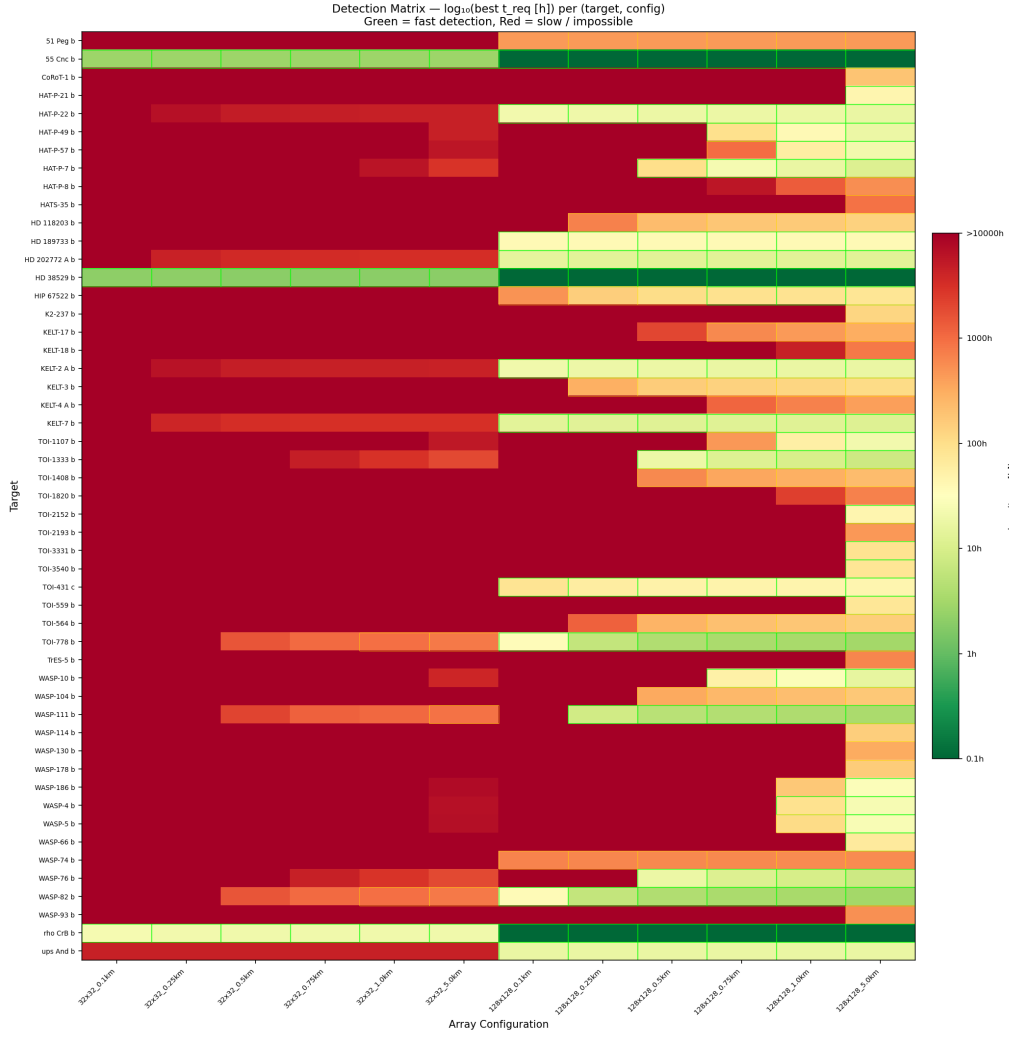


Figure 9: Detection feasibility matrix for the full array-configuration trade-off study. Green: feasible (< 100 h); yellow: aspirational (100–1000 h); red: impossible. Rows are configurations; columns are targets ordered by decreasing flux density.

12. Results and Comparison

12.1. Standard Configuration Comparison

Table 2 summarises the detection yield for all 16 standard configurations across the full 373-target catalogue.

Table 2: Array configuration summary. $N_{\text{det}}(< 100 \text{ h})$ counts targets detectable in $< 100 \text{ h}$ in any frequency band.

Configuration	N_{elem}	B_{max} (km)	HPBW at 30 MHz (arcmin)	$N_{\text{det}}(< 100 \text{ h})$
ring_core32x32_1.0km	1088	2.08	579	–
ring_core32x32_5.0km	1088	10.08	63	–
ring_core128x128_1.0km	17408	2.33	579	–
ring_core128x128_5.0km	17408	10.33	63.8	34
y_shape_core32x32_1.0km	1088	1.81	578	–
y_shape_core32x32_5.0km	1088	8.74	63	–
y_shape_core128x128_1.0km	17408	2.06	578	–
y_shape_core128x128_5.0km	17408	8.99	64.9	33
random_core32x32_1.0km	1088	1.55	576	–
random_core32x32_5.0km	1088	7.41	63	–
random_core128x128_1.0km	17408	1.79	576	–
random_core128x128_5.0km	17408	7.65	64.0	31
line_core32x32_1.0km	1088	2.08	576	–
line_core32x32_5.0km	1088	10.08	63	–
line_core128x128_1.0km	17408	2.33	576	–
line_core128x128_5.0km	17408	10.33	65.0	34

N_{det} for 32×32 and 1 km configurations is < 5 across all bands and omitted.

Key findings:

- The 128×128 core is essential: the 32×32 arrays have insufficient collecting area to detect any target in $< 100 \text{ h}$.
- A 5 km outrigger separation is essential for angular resolution: at 1 km the HPBW at 30 MHz is ~ 580 arcmin, too large to discriminate targets from background.
- The ring and line arrays share the largest B_{max} (10.33 km) and thus the sharpest beam and lowest confusion limit, yielding the most detections (34 each).
- Geometry differences among the four outrigger patterns are small: $\Delta N_{\text{det}} \leq 3$ across ring/Y-shape/random/line.

12.2. Cross Configuration Comparison

Table 3 compares the two cross configurations against the four $128 \times 128 + 5 \text{ km}$ reference arrays.

Table 3: Cross configuration comparison results.

Configuration	N_{elem}	B_{max} (km)	$N_{\text{det}}(< 100 \text{ h})$	MSL (dB)
Ring	17408	10.33	34	−3.8
Y-shape	17408	8.99	33	−3.3
Random	17408	7.65	31	−3.8
Line	17408	10.33	34	−3.4
Cross E-W	17408	8.33	33	− 4.3
Cross N-S	17408	8.33	33	− 4.3

The cross configurations achieve:

- Best MSL (−4.3 dB vs. −3.3 to −3.8 dB for symmetric arrays).
- Competitive detection yield (33 targets) despite a shorter maximum baseline (8.33 vs. 10.33 km for ring and line).
- Elongated UV coverage providing anisotropic resolution, which may be preferred for specific science cases (e.g. mapping extended plasma structures).

12.3. Beam Symmetry Analysis

A dedicated asymmetric beam study was conducted to investigate whether array geometry asymmetry is detectable in the power beam. Arrays with deliberately broken symmetry (line: all outriggers along E-W axis) were compared to rotationally symmetric arrays (ring) at multiple pointing directions.

Conclusion: The power beam $B(l, m) = |\text{AF}|^2$ is always 180° -rotationally symmetric due to the Hermitian symmetry of the Fourier transform. Asymmetry information is contained only in the *phase* of the complex AF, not in the magnitude. No error in the code was found; the symmetry is physically correct.

12.4. Extended Source Response

An extended source analysis was performed using a $32 \times 32 + 5$ km array (chosen for sufficient beam resolution in the numerical grid) to test whether geometry differences become visible for spatially resolved sources.

The beam-weighted flux-recovery fraction:

$$\eta_{\text{BW}} = \frac{\sum_{l,m} I_{\text{true}}(l, m) B(l, m)}{\sum_{l,m} I_{\text{true}}(l, m)} \quad (21)$$

was computed as a function of source angular size and position angle (PA).

Results:

- For point sources, $\eta_{\text{BW}} = 1$ identically for all geometries (as expected).
- For extended sources, η_{BW} varies with source size and PA, and the variation is geometry-dependent.

- The line array shows the largest PA-orientation bias ($\Delta\eta = 0.0111$, the flux-recovery range over all tested PAs).
- The ring array shows the smallest PA-orientation bias ($\Delta\eta = 0.0002$), consistent with its near-isotropic UV coverage.

Since ALO's primary targets are point-like exoplanets (angular size $\ll$ HPBW ≈ 64 arcmin at 30 MHz), extended-source response is not a discriminating metric for the primary science case.

13. Final Conclusions

1. **Core size is the dominant driver of sensitivity.** The 128×128 core (17 408 elements, $A_{\text{eff}} \approx 13,000 \text{ m}^2$ at 30 MHz) is necessary for feasible detections; the 32×32 core is insufficient.
2. **Outrigger separation drives angular resolution.** At 5 km the HPBW at 30 MHz is ~ 64 arcmin; at 1 km it is ~ 578 arcmin. The long baseline suppresses the confusion limit by a factor of 9, enabling detection of targets that would otherwise be confusion-dominated.
3. **Outrigger geometry has a second-order effect on detection yield.** Differences in feasible detection count among ring, Y-shape, random, and line are only 1–3 targets out of 34. The choice of geometry should be driven by other considerations (e.g. UV-coverage isotropy, sidelobe characteristics, construction constraints).
4. **Cross configurations offer the best sidelobe suppression.** Irregular arm lengths (1/1/5/3 km) break the periodicity that causes coherent aliasing rings, reducing MSL by 0.5 dB relative to the best symmetric array. This comes at the cost of a slightly shorter maximum baseline (8.33 vs. 10.33 km) and hence one fewer feasible detection.
5. **The 20–40 MHz sub-band is the most productive.** Most feasible detections occur in this band due to a combination of lower sky temperature, higher predicted ECM flux, and sufficient angular resolution at $B_{\text{max}} = 8\text{--}10$ km.
6. **Point-source sensitivity is geometry-independent.** The power beam $|AF|^2$ is Hermitian-symmetric for any real-weight array. Geometry information in $|AF|^2$ is limited to sidelobe structure; main-lobe solid angle depends only on the total core aperture, not on outrigger pattern. This is a fundamental physical result, not a modelling limitation.

14. Recommendations for Next Steps

1. **Beam forming with weights.** Replace unit weights with an apodisation taper (e.g. Hanning window) to reduce sidelobes further. A Taylor taper with $n = 5$ sidelines could reduce MSL to ~ -20 dB at the cost of $\sim 20\%$ wider main lobe.
2. **Realistic beam model including element pattern.** Include the element (dipole) radiation pattern $E(\theta, \phi)$ as a multiplicative factor in the AF. This will create an element-pattern null towards the horizon and modify the effective collecting area at large zenith angles.

3. **Lunar rotation and source tracking.** Compute the UV-coverage track for a full ~ 14 -day lunar day, including diurnal rotation of the far-side frame. Earth-facing sources appear above the lunar horizon for ~ 11 h per sidereal day and their UV tracks will fill a much larger region of the UV plane.
4. **Optimal sub-array size for cross configuration.** Optimise the outrigger sub-array size N_o for the cross geometry. Larger outriggers increase baseline sensitivity but increase mass and deployment cost.
5. **Calibration architecture.** Design a calibration strategy for the far-side environment. Bright background sources (e.g. Cas A, Cyg A, Vir A) will be visible above the lunar horizon; assess their usability as calibrators for both gain and phase solutions.
6. **Ionosphere-free mode.** Below ~ 1 MHz the Moon's plasma environment (exosphere) may introduce dispersive delays. Extend the sensitivity model to account for this chromatic phase noise.
7. **Comparison with FARSIDE/DAPPER concepts.** Benchmark ALO performance against existing lunar-farside array proposals (FARSIDE, DAPPER, DSL) in terms of collecting area, angular resolution, and detection yield for the same target catalogue.
8. **Deployment feasibility.** Feed the optimal configuration parameters (core size 128×128 , outrigger distance 5 km, cross or ring geometry) into an engineering trade study covering deployment mass, power budget (solar/RTG), and thermal management on the lunar surface.

Appendix: Physical Constants and Parameters

Symbol	Value	Description
d	5.15 m	Dipole element spacing
η	0.70	Aperture efficiency
N_σ	5	Detection significance threshold
N_{pol}	2	Number of polarisations
c	$3 \times 10^8 \text{ m s}^{-1}$	Speed of light
k_B	$1.381 \times 10^{-23} \text{ J K}^{-1}$	Boltzmann constant
φ	-20.38°	Tsiolkovsky crater latitude
λ_{site}	128.97°	Tsiolkovsky crater longitude

Software. All computations were performed using Python 3 with NumPy, SciPy, and Matplotlib. The simulation code is version-controlled in the Git repository `mahshadrashidi/ALO_array_model` on GitHub. All figures in this report are generated directly from the simulation outputs and are reproducible by running `alo_geometry_tradeoff.py`.